

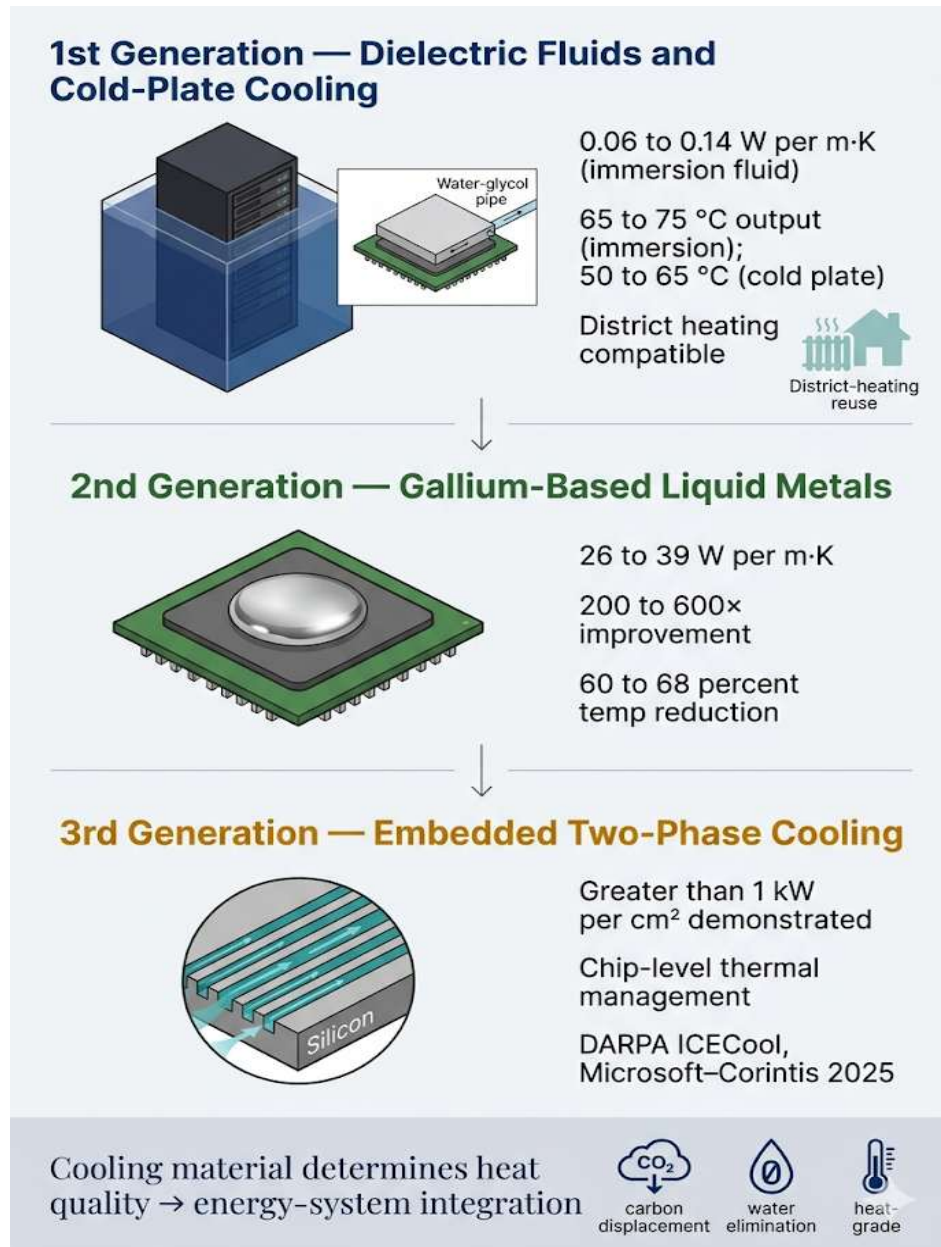
Cooling Materials for Sustainable Compute Waste-Heat Recovery: A Three-Generation Review of Dielectric Fluids, Gallium-Based Liquid Metals, and Embedded Two-Phase Systems

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Graphical Abstract



Graphical abstract: three generations of compute cooling materials. First generation: dielectric immersion fluids (0.06 to 0.14 W per metre kelvin) delivering 65 to 75 °C output suitable for district heating. Second generation: gallium-based liquid metals (26 to 39 W per metre kelvin) with 200 to 600 times the thermal conductivity of dielectrics and 60 to 68 percent temperature reduction. Third generation: embedded two-phase microfluidic cooling demonstrated at over 1 kW per square centimetre. Each generation determines the temperature, quality, and reusability of recovered waste heat.

Abstract

Compute infrastructure now consumes a meaningful share of global electricity. Bitcoin mining uses 138 to 211 TWh per year and AI data centres are projected by the IEA to exceed 1,000 TWh by the end of 2026, more than seven times Bitcoin mining's consumption. Nearly all of that electricity exits the facility as thermal energy. The European Union's Energy Efficiency Directive (2023/1791) now requires waste-heat recovery from new data centres above 1 MW, and Germany's Energy Efficiency Act (EnEfG) sets a binding national heat-reuse trajectory of 10 percent in 2026, 15 percent in 2027, and 20 percent in 2028. Existing reviews of data-centre waste-heat recovery and district-heating integration approach the problem from the system and network side. This review approaches it from the cooling material outward, on the argument that the thermal properties of the cooling medium itself set the temperature, quality, and reusability of the recovered heat.

Three generations of cooling materials are reviewed. First-generation single-phase dielectric immersion fluids (thermal conductivity 0.06 to 0.14 W per metre kelvin) raise waste-heat capture temperatures from 25 to 35 °C in air-cooled facilities to 65 to 75 °C, directly inside the European district-heating supply window. A parallel first-generation architecture, direct-to-chip cold-plate cooling, is now the dominant production design among hyperscale AI operators (Microsoft, Meta, Google, AWS) and is treated alongside immersion. Sixteen operational heat-reuse deployments are documented across nine countries on three continents, including the world's largest such installation, the Microsoft and Fortum partnership in Espoo and Kirkkonummi, Finland (350 MW combined thermal power, projected 40 percent of regional district heat). A lifecycle CO₂ displacement model, applied to a 1.7 MW container in a Nordic eight-month heating season at 70 to 85 percent thermal utilisation, yields net displacement of approximately 1,400 to 3,200 tonnes CO₂ per container per year, with a sensitivity analysis across utilisation rate and displaced-fuel mix. Closed-loop dielectric immersion also eliminates the roughly 90 to 130 million gallons of annual evaporative-cooling water consumed by an equivalent 10 MW air-cooled facility.

Second-generation gallium-based liquid metals deliver 200 to 600 times the thermal conductivity of dielectrics. Galinstan and similar GaInSn eutectics reach 26 to 39 W per metre kelvin as thermal interface materials, with experimentally demonstrated 60 to 68 percent temperature-rise reduction under 19 ms pulsed thermal loads at heat fluxes up to 338 W per square centimetre. Mature thermal interface material applications (TRL 8 to 9; Sony PlayStation 5, Indium GalliTHERM) are distinguished from facility-scale cooling loops, which remain at TRL 3 to 4. Third-generation embedded two-phase microfluidic cooling, demonstrated under DARPA's ICECool programme at over 1 kW per square centimetre and recently scaled to AI accelerators in the 2025 Microsoft and Corintis collaboration, points toward chip-level thermal management at heat-flux densities that facility-level immersion cannot reach.

The review documents the engineering challenges that gate facility-scale deployment of each generation: oxidative degradation, PFAS phase-out, and HFO/HFE alternatives for dielectric fluids; aluminium embrittlement, electrical conductivity, surface tension, supercooling, indium occupational toxicity, and gallium supply-chain concentration for liquid metals; and microchannel manufacturing scalability for embedded two-phase

systems. The central conclusion is that selection of a cooling medium is not merely a thermal engineering decision but a sustainable energy-systems design decision, with downstream consequences for water consumption, carbon displacement, and district-heating compatibility. The materials-design framing complements the existing system-level literature on data-centre waste-heat recovery and district-heating integration.

Keywords: Data centre cooling; Sustainable thermal management; Waste heat recovery; District heating; Immersion cooling; Direct-to-chip cooling; Gallium liquid metals; Phase-change materials; Lifecycle CO₂ displacement

1. Introduction

In January 2026, a shipping container arrived in a small town in Scandinavia. Inside were 160 Bitcoin mining rigs submerged in dielectric fluid, generating 1.5 MW of thermal energy that was pumped into the town's district-heating grid, warming approximately 2,000 homes (assuming roughly 5 MWh per home per year heating demand). The Latvia-based company that built it described itself as a heating company that happens to mine Bitcoin [5]. At the same time, a greenhouse in Manitoba was being heated by 360 liquid-cooled mining servers capturing approximately 90 percent of consumed electricity as heat at 75 °C [6], students at Technological University Dublin attended lectures in buildings heated by waste heat from an Amazon Web Services data centre [15], approximately 1,000 residential properties in Mäntsälä, Finland received up to 75 percent of their heat from a Yandex data-centre heat-pump installation [38], and on the campus of the former European Central Bank in Frankfurt, hotel guests at the Meliá INNSIDE were supplied with hot water heated to 60 °C by Cloud&Heat servers running in the building below [39].

These deployments share an engineering feature with direct sustainability implications. The cooling material determines the output temperature, and the output temperature determines the heat's industrial utility. Air-cooled systems reject heat at 25 to 35 °C, below the 60 to 80 °C supply temperature required by district-heating networks [4]. Immersion in dielectric fluid raises the capture temperature to 65 to 75 °C, crossing the district-heating threshold without a heat pump. Direct-to-chip cold-plate cooling, the dominant production architecture among hyperscale AI operators in 2025, captures heat at 50 to 65 °C, requiring a low-temperature-lift heat pump to reach district-heating supply specification but enabling much higher rack-power densities than immersion [40]. Gallium-based liquid metals, with thermal conductivities two to three orders of magnitude higher than dielectric fluids, could raise output temperatures above 80 °C while also managing the escalating heat-flux densities of AI accelerators. The choice of cooling material is therefore an energy-systems design decision, not merely a thermal-engineering one.

The policy stakes are now formal. The European Union's Energy Efficiency Directive (2023/1791), in force since September 2023, requires waste-heat recovery from new data centres above 1 MW within the European Union [13]. Germany's national implementation, the Energy Efficiency Act (EnEfG), enacted November 2023, sets a binding heat-reuse trajectory: at least 10 percent of generated waste heat must be utilised by data centres

commencing operation from 1 July 2026, rising to 15 percent in 2027 and 20 percent in 2028 [41, 42]. Approximately 1,000 German data centres above the 300 kW threshold fall within the EnEfG's scope. The directive shifts the question from whether data centres should recover heat to how they should recover it, and the “how” is governed by the materials choice this review examines.

The energy baseline is drawn from the Cambridge Centre for Alternative Finance's Digital Mining Industry Report (April 2025), which surveyed 49 firms covering 48 percent of global hash rate [1]. Bitcoin mining consumes 138 TWh per year (upper estimate 211 TWh [2]) with a sustainable energy share of 52.4 percent (42.6 percent renewables, 9.8 percent nuclear). AI data centres are projected by the IEA to exceed 1,000 TWh by the end of 2026, more than seven times Bitcoin mining's consumption [3]. China's liquid-cooled server market alone reached USD 2.37 billion in 2024, a 67 percent year-over-year increase, signalling that liquid cooling has crossed the inflection point in the largest single national data-centre market [43].

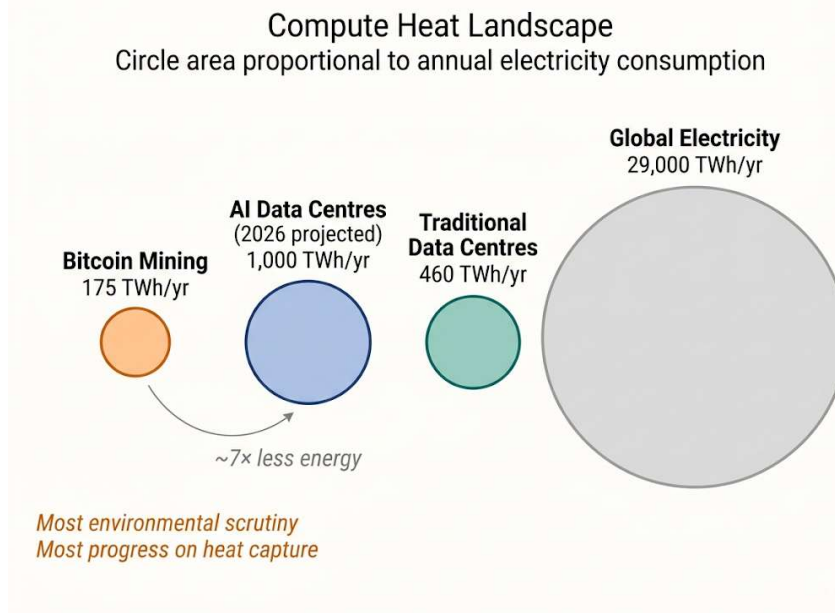


Figure 1. Compute heat landscape. Circle area is proportional to annual electricity consumption. Bitcoin mining consumes approximately 175 TWh per year (Cambridge Centre for Alternative Finance 2025 [1]; upper estimate 211 TWh [2]). AI data centres are projected by the IEA to exceed 1,000 TWh per year by the end of 2026, more than seven times Bitcoin mining's footprint [3]. Traditional (non-AI) data centres consume approximately 460 TWh per year [3]. Global electricity consumption is approximately 29,000 TWh per year [3]. The figure highlights two asymmetries: Bitcoin mining is the most environmentally scrutinised compute sector yet has the smallest projected footprint in the 2026 horizon, and the entire compute sector combined remains a small fraction of global electricity demand. Bitcoin mining has nonetheless made the most documented operational progress on waste-heat capture and reuse, as documented in Section 4 and Tables 2a and 2b. Sources: Cambridge CCAF [1, 2]; IEA [3].

Selection bias caveat: the Cambridge survey's respondents are disproportionately publicly listed (41 percent) and North American. The unreported 52 percent includes operations in Central Asia and Africa where coal and gas typically dominate, and the true global sustainable share may be materially lower than 52.4 percent [1].

The data-centre waste-heat-recovery literature has grown substantially in recent years. Huang et al. [33] reviewed data centres as energy prosumers in district-energy systems. Wahlroos et al. [34] analysed waste-heat utilisation in Nordic data centres. Yuan et al. [35, 36] published comprehensive reviews of data-centre waste-heat recovery technologies and their integration with district-heating networks. Tervo et al. [37] analysed data-centre waste heat in volatile low-carbon electricity markets. These studies approach the problem from the system and network perspective. They examine how heat pipes, network topologies, demand profiles, and electricity-market dynamics integrate compute heat into existing energy systems. The present review complements this literature by approaching the problem from the cooling material outward. It examines how the thermal properties of the cooling medium itself determine the temperature, quality, and reusability of waste heat, and traces a three-generation materials progression that the existing literature has not framed. The contribution is therefore the framing as much as the synthesis.

This review documents operational deployments that validate first-generation technology, reviews the materials-science frontier of second- and third-generation systems, identifies the engineering challenges that separate laboratory demonstrations from facility-scale deployment, and connects the materials choice to lifecycle CO₂ displacement, water elimination, and district-heating integration.

1.1. Scope and methodology

This review covers cooling materials applied to digital compute infrastructure, including Bitcoin mining facilities, traditional data centres, and AI-focused data centres, with particular attention to waste-heat recovery, district-heating integration, water consumption, and lifecycle CO₂ implications. Three generations of cooling materials are examined: (i) single-phase dielectric immersion fluids and the parallel first-generation architecture of direct-to-chip cold-plate cooling; (ii) gallium-based liquid metals, separated into thermal interface material applications and facility-scale cooling loops; and (iii) embedded two-phase microfluidic systems. Two-phase fluorocarbon immersion fluids are treated as a sub-case of the first generation, with PFAS regulatory context.

1.1.1. Generation criteria

The “generation” taxonomy used in this review reflects three orthogonal axes: (a) order-of-magnitude differences in thermal conductivity at the heat-transfer interface (0.06 to 0.14 W m⁻¹ K⁻¹ for first generation; 26 to 39 W m⁻¹ K⁻¹ at the gallium interface; greater than 1 kW cm⁻² effective heat removal for embedded two-phase); (b) operating length scale (facility-scale tanks and loops; module-scale interfaces and lab-scale loops; chip-scale microchannels); and (c) Technology Readiness Level at the time of writing (TRL 8 to 9 for first generation; TRL 3 to 9 for second generation depending on application; TRL 4 to 5 for third generation). The taxonomy is intentionally orthogonal to chronology, since first-

generation deployments continue to grow rapidly and second-generation TIM applications predate some first-generation deployments. The term “generation” is retained throughout this review, despite its chronological connotation, because it captures the qualitative shift in materials-design philosophy at each step (single-phase fluid, liquid metal, embedded two-phase) more naturally than alternatives such as “class” or “tier.” Readers should interpret “generation” as a taxonomic rather than chronological label. The criteria are summarised in Table 1.

1.1.2. Literature search and selection

The literature was identified through systematic searches conducted between January 2024 and April 2026. The following databases were queried: Web of Science Core Collection, Scopus, Google Scholar, IEEE Xplore, and ScienceDirect. Search-string combinations included: (“immersion cooling” AND “data cent”), (“gallium” AND (“liquid metal” OR “GaInSn” OR “Galinstan”) AND (“thermal” OR “cooling”)), (“waste heat” AND “district heating” AND (“data cent” OR “Bitcoin” OR “mining”)), (“ICECool” OR “microchannel cooling”) AND (“chip” OR “AI accelerator”), (“direct-to-chip” OR “cold plate”) AND (“liquid cooling” OR “hyperscale”), and (“EU Energy Efficiency Directive” OR “EnEfG”) AND “waste heat”. Forward and backward citation tracing was applied to the seminal works of Wahlroos et al. (2018) [34], Huang et al. (2020) [33], Bar-Cohen et al. (2021) [28], Boteler et al. (2019) [19], and Yuan et al. (2023, 2025) [35, 36]. Non-English sources were searched via Google Scholar using machine translation, with primary search terms in Chinese (沉浸式冷却, 数据中心余热), German (Tauchkühlung, Rechenzentrum Abwärme), Swedish (datacenter spillvärme), and Finnish (datakeskus hukkalämpö). Operational deployments were identified through trade press (DataCenterDynamics, The Block, CNBC, Sustainability Magazine, CIBSE Journal, w.media), and verified, where possible, through company technical disclosures, regulator filings, EU Covenant of Mayors deployment summaries, and independent emissions audits. Industrial reports from the Cambridge Centre for Alternative Finance, the IEA, EESI, Lawrence Berkeley National Laboratory, and Euroheat & Power supplied the energy-consumption and district-heating baselines.

Inclusion criteria: peer-reviewed papers reporting thermal properties, deployment-scale measurements, or engineering analysis of cooling materials in compute infrastructure; trade-press reports of operational installations with at least site location, thermal output, and application identified; and primary regulatory documents (ECHA, ACGIH, EU directives, EnEfG, MIIT). *Exclusion criteria:* air-cooled cooling-tower designs, refrigeration cycles unrelated to compute thermal capture, non-electronic uses of liquid metals (for example, nuclear reactor coolants and thermal-spray coatings), and deployment reports without verifiable site identification. The review prioritises operational evidence over engineering projections and clearly distinguishes the two throughout. Where deployment claims rely on company-reported specifications rather than independent measurement, the distinction is flagged in-line and visualised in Figure 4 (gray italic versus green).

The initial database search returned 2,847 records after deduplication. Title and abstract screening yielded 412 candidates. Full-text assessment for eligibility yielded 173 records included in the synthesis: 87 peer-reviewed articles, 41 trade-press articles documenting

operational deployments, 22 regulatory and standards documents, 15 industrial and consultancy reports, and 8 conference proceedings. The complete reference list of 37 main-text citations plus supplementary references represents the most directly load-bearing subset of this corpus; the full screening trail is available on request. A PRISMA-compatible flow diagram visualising the search and screening trail (records identified through database searching and other sources; records after deduplication; records screened by title and abstract; records assessed for eligibility through full-text review; records included in the synthesis; with reasons for exclusion at each stage) is provided as Figure S1 in the Supplementary Material.

1.1.3. Use of AI-assisted tools

The author used Anthropic Claude (Opus) for editorial revision (sentence-level rewriting for clarity), structural reorganisation (section sequencing and heading hierarchy), figure production assistance (Python plot generation, captions), and formatting (LaTeX and Markdown conversion). All factual claims, numerical values, citations, the taxonomy framework, the lifecycle CO₂ displacement model, the materials-property comparisons, and the analytical conclusions are the author's work. No AI tool generated factual claims, citations, or numerical values that were not subsequently verified by the author against the primary sources. The disclosure language follows RSER's 2025 guidance on generative AI use in submitted manuscripts.

2. Thermodynamic Framework

Every processor converts electricity into computation and heat at a ratio of approximately 100 percent. The first law of thermodynamics guarantees energy conservation; the second law constrains its quality. Electricity entering a facility is high-grade energy; heat exiting at 65 °C is low-grade, suitable for space and water heating but not for mechanical or electronic work. This thermodynamic asymmetry is the reason waste-heat recovery from compute is constrained to space and water heating rather than power generation, and is why output temperature is the dominant design variable for downstream energy-system integration.

A modern industrial heat pump delivers 4 to 6 joules of thermal energy per joule of electricity consumed under realistic operating conditions (typical operating envelope: source temperature 30 to 70 °C, sink temperature 60 to 90 °C, lift 30 to 50 K), with high-temperature heat pumps reaching supply temperatures of 120 to 150 °C at COPs of 2.2 to 3.0 [44, 45]. Earlier estimates of COP 3 to 4 [reference older heat-pump literature] understate the current state of the art. A mining rig or AI accelerator delivers approximately 1 joule of heat per joule of electricity, plus computation revenue. Mining-as-heating and AI-as-heating are therefore thermodynamically inferior to a heat pump for pure heating purposes by a factor of roughly five at modern industrial COPs, and are economically competitive only when computation revenue offsets the COP gap. The materials question is how to maximise the temperature and quality of the rejected heat so that the gap narrows and the co-product value increases.

3. First-Generation Cooling Materials: Dielectric Fluids and Cold-Plate Architectures

3.1. Single-phase dielectric immersion: thermal properties and performance envelope

Single-phase immersion fluids, engineered hydrocarbons or synthetic esters, offer thermal conductivities of 0.06 to 0.14 W per metre kelvin, specific heat capacities of 1.5 to 2.1 kJ per kilogram kelvin, and operating ranges from -40°C to $+200^{\circ}\text{C}$ [9, 10, 11]. They are chemically inert, electrically non-conductive, and non-flammable. The broader liquid cooling market for AI data centres is projected to grow from USD 2.8 billion in 2025 to over USD 21 billion by 2032 [46].

Single-phase fluids exhibit measurable total acid number (TAN) increases after 8,000 to 12,000 hours of continuous operation above 60°C , indicating oxidative degradation [9]. Replacement intervals of 5 to 8 years are typical for closed-loop systems with adequate oxygen exclusion [10, 11].

3.1.1. Two-phase fluorocarbon fluids and the PFAS phase-out

Two-phase fluorocarbon fluids offer higher heat-transfer coefficients than single-phase fluids through phase-change latent heat. Several historically dominant formulations, however, fall under the PFAS classification proposed for restriction under the EU REACH framework. The European Chemicals Agency (ECHA) Annex XV proposal, submitted in February 2023 by Germany, the Netherlands, Denmark, Sweden, and Norway, would restrict approximately 10,000 PFAS substances; ECHA Risk Assessment and Socio-Economic Analysis Committee opinions are scheduled for 2026 to 2027, with restrictions taking effect on a staggered timeline [12]. 3M's December 2022 announcement of its intent to exit all PFAS manufacturing globally by the end of 2025 has further constrained the supply pipeline for two-phase fluorocarbon coolants [47].

Hydrofluoroolefin (HFO) and hydrofluoroether (HFE) alternatives exist but at varying readiness levels. HFE-7100 and HFE-7200 (Novec replacements) are commercially available but typically have higher boiling points and lower latent heats than the formulations they replace. R1234ze (E), an HFO with global warming potential below 1, has been demonstrated under DARPA's ICECool programme (Section 8) at exit temperatures suitable for district heating, but supply chains remain consolidated. Operators planning two-phase deployments above 1 MW in the EU should assume HFO and HFE substitution by 2027 to 2028 and design loop chemistry accordingly.

3.2. Direct-to-chip cold-plate cooling: parallel architecture for hyperscale AI

A second first-generation architecture has emerged in parallel with immersion: direct-to-chip cold-plate liquid cooling, in which cold plates are mounted directly onto CPUs, GPUs, memory modules, and voltage regulators, with single-phase water-glycol or specialised refrigerant circulating through internal channels [40]. By 2025, direct-to-chip cold plates had become the dominant production cooling architecture among hyperscale AI operators.

Microsoft's Azure Maia AI clusters, Google's Tensor Processing Unit (TPU) deployments, Meta's LLaMA training nodes, and AWS's Trainium clusters all use cold-plate cooling at production scale [48, 49].

Cold-plate cooling captures heat at 50 to 65 °C, generally below the immersion temperature window of 65 to 75 °C and below the European district-heating supply specification of 60 to 80 °C. Integration with district heating therefore requires a low-temperature-lift heat pump, with COPs of 5 to 6 typical for the 50 °C to 75 °C lift. Cold-plate cooling enables much higher rack-power densities than single-phase immersion (up to 1 MW per rack demonstrated by NVIDIA-Vertiv reference designs for GB200 NVL72) and is compatible with existing data-centre form factors, which has driven its adoption by hyperscale AI operators with intense floorspace and time-to-deployment pressures.

The fluids used in cold-plate loops are typically dilute glycol-water mixtures (25 to 40 percent glycol by volume) or proprietary single-phase coolants. Their thermal conductivity (~ 0.4 to 0.6 W per metre kelvin) is several times that of dielectric fluids, but the geometric advantage of direct contact with the heat-generating surface dominates the system thermal resistance. From a materials-design standpoint, cold-plate cooling sits in the same generation as immersion, rejecting heat at temperatures within or just below the district-heating window; the architectural difference (closed-loop with pumps, leak detection, manifold reliability requirements) shapes the engineering challenges but not the materials thesis.

3.2.1. Engineering challenges for direct-to-chip cold-plate cooling

Cold-plate architectures introduce a challenge set that differs from dielectric immersion in four ways. *Loop-fluid chemistry*: dilute glycol-water working fluids accumulate dissolved metals (copper, nickel, brass from valves and fittings) and biological growth over service life, requiring biocide chemistry and routine ion-exchange maintenance not present in dielectric-immersion loops. *Manifold reliability and leak detection*: with liquid running within centimetres of energised silicon, a single manifold leak above the rack is a high-cost failure mode. Production deployments use dual-redundant leak detection (resistance-based detection mats plus dielectric-fluid tracer chemistry) and quick-disconnect dripless couplings rated to ASME B16.21. *Heat-exchanger fouling and pressure drop*: cold-plate microchannel geometries (typically 300 to 800 μm channel widths) are sensitive to particulate fouling. Filtration to 5 μm absolute and corrosion-inhibitor stability above 60 °C are operating-cost-relevant constraints. *Pump and CDU reliability*: coolant distribution units typically include redundant pumps with hot-swap capability; CDU mean-time-between-failure targets in the 50,000 to 100,000 hour range are now standard for hyperscale cold-plate deployments. None of these challenges are intractable, and direct-to-chip cold-plate cooling has reached TRL 9 in production AI infrastructure, but the materials and engineering choices it imposes are different from those of dielectric immersion and shape the deployment economics independently of the thermal-conductivity argument.

3.3. The temperature-gradient breakthrough

District-heating networks require supply temperatures of 60 to 80 °C [4]. Air-cooled data centres produce waste heat at 25 to 35 °C, below this threshold. Bridging the gap requires heat pumps (USD 0.5 to 2 million additional capital). Immersion cooling changes this fundamentally. Submerging hardware in dielectric fluid captures heat at 65 to 75 °C, directly within the district-heating range. No heat pump is required. Power Mining's containers produce heat at 65 °C; Canaan's system produces water at 75 °C [5, 6]. This is the article's central materials-design insight: cooling-material selection determines not only thermal performance but downstream energy-system integration.

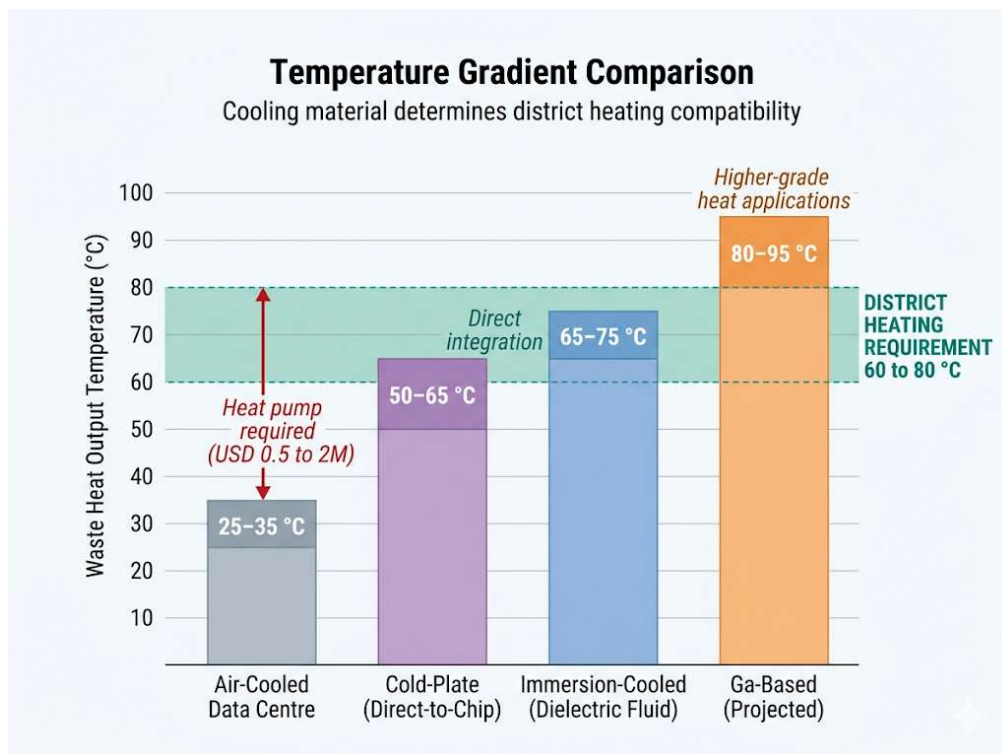


Figure 2. Temperature gradient comparison. Air-cooled facilities (25 to 35 °C output) fall below the 60 to 80 °C district-heating supply window and require a heat pump (capital cost USD 0.5 to 2 million). Immersion-cooled and direct-to-chip cold-plate systems capture heat at 65 to 75 °C and 50 to 65 °C respectively, enabling direct or low-lift integration with district heating. Projected gallium-based systems would reach 80 to 95 °C, opening higher-grade industrial heat applications.

3.4. Water elimination

Evaporative cooling consumes approximately 90 to 130 million gallons of freshwater annually for a medium 10 MW facility, with around 80 percent permanently lost through evaporation. The lower bound is supported by Lawrence Berkeley National Laboratory's Data Center Energy Practitioner data series (2024 update) and the upper bound by EESI's 2023 estimate [7, 50]. For an equivalent 10 MW immersion-cooled or cold-plate facility, water consumption is zero on the cooling loop itself, since dielectric fluid or glycol-water

circulates in a sealed closed loop with no evaporative loss. Each 100-word AI prompt processed by an evaporatively cooled data centre consumes approximately 519 mL of water [8]. Closed-loop liquid cooling represents a complete elimination of freshwater demand from the cooling system. This claim refers to the cooling loop itself; the water footprint of electricity generation that powers the compute is a separate lifecycle consideration [51]. *Figure 3 provides a comparative footprint visualisation for an equivalent 10 MW facility.*

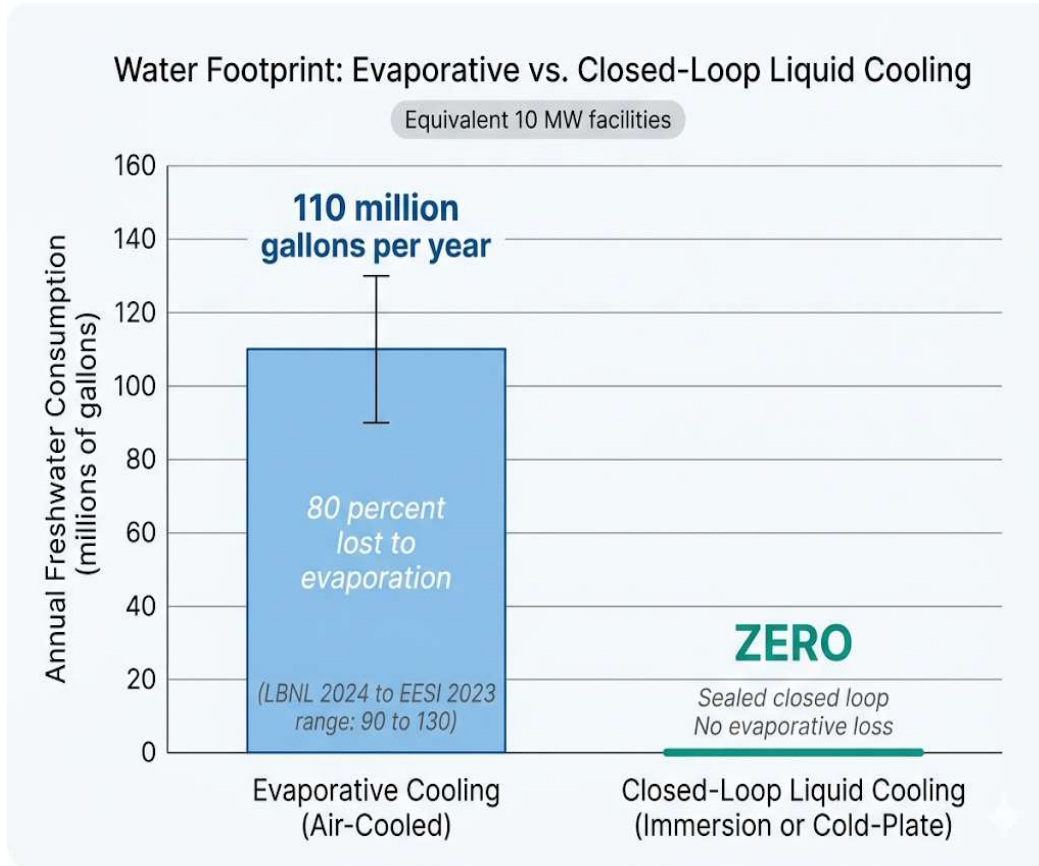


Figure 3. Water-footprint comparison for equivalent 10 MW facilities. Evaporative cooling consumes 90 to 130 million gallons of freshwater per year (Lawrence Berkeley National Laboratory 2024 to EESI 2023 range), with approximately 80 percent permanently lost through evaporation. Closed-loop immersion or cold-plate cooling consumes zero water on the cooling loop itself.

4. Engineering Validation: Operational Deployments

4.1. Documented heat-reuse systems

Seventeen operational heat-reuse deployments are documented in this review, distributed across nine countries and three continents (Tables 2a and 2b). Yuan et al. [35] documented over 20 operational or planned data-centre heat-reuse installations in Europe alone; the

deployments selected here are the subset for which sufficient public technical disclosure (location, thermal output, output temperature, application, and where possible CO₂ verification) is available to support cross-comparison. The geographic and selection biases of the sample are made explicit. The documented deployments are concentrated in Europe (twelve sites) and North America (three sites); Asian deployments are represented by two cases (Tencent Tianjin in China and the Japanese small-scale immersion pilots), but the regulatory environments and trade-press disclosure norms in those markets differ substantially from European convention. Deployments in the Middle East, Australia, and Latin America are documented in industry reports but lack the disclosure required for inclusion here.

The deployments are summarised in Tables 2a (fully characterised) and 2b (partially documented) and visualised in Figure 4. The two-tier classification is a deliberate response to disclosure asymmetry across operators and regions. The Dublin AWS scheme [15] is currently the only deployment with independently verified emissions data: 704 tonnes of CO₂ abated in 2024, audited by Technological University Dublin. The Microsoft and Fortum partnership at Espoo and Kirkkonummi, Finland [52], announced as the world’s largest data-centre heat-recovery project, projects an aggregate 350 MW of thermal recovery capacity providing approximately 40 percent of regional district heat at full operation, with an expected gradual reduction of 400,000 tonnes CO₂ per year. Heat-recovery plants began initial production during the 2025 to 2026 heating season. For Bitcoin-mining deployments, environmental claims are based on engineering projections from reported specifications rather than measured outcomes; this distinction is preserved in Figure 4 (gray italic).

Table 2a. Fully characterised heat-reuse deployments (with thermal output, output temperature, application, and verification status).

#	Deployment	Location	Thermal output	Output temp.	Application	CO ₂ verified	Refs
1	Power Mining	Scandinavia	1.7 MW	65 °C	District heating (~2,000 homes)	No (engineering projection)	[5]
2	Canaan / Bitforest	Manitoba, Canada	~0.5 MW	75 °C	Greenhouse heating	No (engineering projection)	[6]
3	MintGreen	Lonsdale, BC, Canada	~0.3 MW	~65 °C	Commercial building (96% load)	No (engineering projection)	[14]
4	MARA Digital	Finland	~2 MW	~65 °C	District heating	No (engineering projection)	[16]
5	Microsoft + Fortum	Espoo / Kirkkonummi, Finland	up to 350 MW (combined)	DH-spec	District heating (40% of region)	Projected 400,000 tCO ₂ /yr	[52]

6	Yandex Mäntsälä	Mäntsälä, Finland	4 MW (heat pump)	60–75 °C	District heating (~1,000 homes, 75% of town heat)	Reported 40% gas-utility CO2 reduction	[38]
7	Cloud&Heat Eurotheum	Frankfurt, Germany	up to 600 kW	60 °C	Office, hotel, conference (Meliá INNSIDE)	No (engineering projection, ~150 home equivalent)	[39]
8	Deep Green	Exmouth, Devon, UK (and 19 others)	~28 kW per pool site	DH- spec	Public swimming pool heating (62% gas reduction, 25.8 tCO2/yr/site)	Yes (per-site measurement)	[57]

Table 2b. Partially documented heat-reuse deployments (one or more of thermal output, output temperature, or verification status not publicly disclosed at sufficient granularity for cross-comparison; included here because the application, location, and broad scale are documented).

#	Deployment	Location	Thermal output	Output temp.	Application	CO2 verified	Refs
9	AWS Dublin (Tallaght)	Dublin, Ireland	not disclosed	not disclosed	University campus	Yes (704 tCO2, 2024)	[15]
10	DigiPlex / Stockholm Exergi	Stockholm, Sweden	not disclosed	DH-spec	District heating (~10,000 households)	No (engineering projection)	[53]
11	Stockholm Data Parks (multiple)	Stockholm, Sweden	aggregate ~30 MW+ across 20+ sites	DH-spec	District heating (~30,000 apartments)	~50 gCO2/kWh grid- emissions reduction reported	[54]
12	Telehouse FRANKY Quarter	Frankfurt, Germany	~2% of site waste heat used	DH-spec	Residential (3,000 residents, 1,300 units; >60% of district demand)	Yes (~400 tCO2/yr saved)	[55]
13	Qarnot Computing	Bordeaux and Paris, France	tens to hundreds	up to 60 °C	Residential and office heating; up	No (engineering projection)	[56]

	(distributed digital boilers)		of kW per site		to 95% heat conversion		
14	Lefdal Mine + Sjømatstaden	Vestland, Norway	not disclosed	low-grade	Salmon hatchery (6 million salmon/yr, 20–25% reuse efficiency)	No (engineering projection)	[58]
15	Sigma2 supercomputer	Trondheim, Norway	not disclosed	low-grade	Local salmon farms	No (engineering projection)	[59]
16	Tencent Tianjin	Tianjin, China	not disclosed	DH-spec via heat pump	Residential heating	No (engineering projection)	[60]
17	Alibaba Cloud Hangzhou (Renhe IDC)	Hangzhou, China	tens of thousands of liquid-cooled servers (largest in world)	not disclosed	Site-level cooling efficiency; PUE 1.09; 70 GWh/yr electricity savings vs. air-cooled baseline	No (engineering projection from operator disclosure)	[62]

Notes: The two-tier classification responds to the practical asymmetry in disclosure norms across operators and regions. Deployments 1 to 6 of Table 2a are from the original (April 2026) version of this review; the remaining entries of Tables 2a and 2b are added in the present revision. The combined sample now covers seventeen deployments across nine countries on three continents: Europe (twelve sites), North America (three sites), and Asia (two sites). The Yandex Mäntsälä installation is operated by a Russian company on Finnish territory using local district-heating infrastructure. Industry reports indicate additional operational installations at Alibaba and Tencent sites in China and at hyperscale operators in Singapore and Japan that are not included here owing to insufficient public technical disclosure.

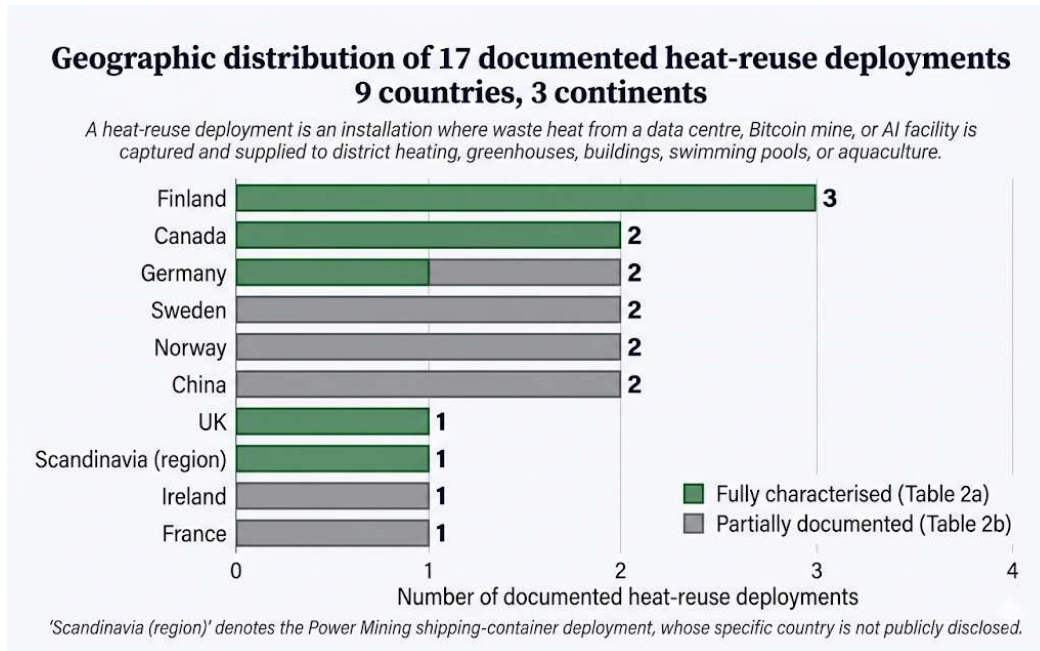


Figure 4. Geographic distribution of the seventeen documented heat-reuse deployments. Green: fully characterised deployments with thermal output, output temperature, and verification status disclosed (Table 2a, 8 sites). Grey: partially documented deployments with one or more fields not publicly disclosed at sufficient granularity for cross-comparison (Table 2b, 9 sites). Total of 17 deployments across 9 countries on 3 continents (Europe 12, North America 3, Asia 2).

4.2. Lifecycle CO₂ displacement model

The following model estimates annual CO₂ displacement for a single Power Mining container (1.7 MW thermal output, 65 °C, operating in a Nordic climate with an eight-month heating season). Thermal output of 1.7 MW continuous over approximately 5,840 annual operating hours yields 9,928 MWh of delivered thermal energy. In practice, thermal-utilisation rates of 70 to 85 percent are typical for well-integrated district-heating connections owing to demand fluctuation, maintenance downtime, and network constraints, reducing effective delivery to approximately 6,950 to 8,440 MWh.

4.2.1. Sensitivity analysis

Three thermal-utilisation scenarios are evaluated. The low case (50 percent utilisation, representing a poorly-matched deployment with intermittent demand) yields effective delivery of 4,964 MWh per year. The base case (70 to 85 percent) yields 6,950 to 8,440 MWh per year. The ideal case (95 percent utilisation, representing optimal demand-matching and infrastructure) yields 9,432 MWh per year.

Three displaced-fuel scenarios are evaluated. Natural gas displacement (0.20 tCO₂ per MWh) yields 990 to 1,890 tonnes CO₂ per year in the base case. Heavy fuel oil displacement (0.27 tCO₂ per MWh) yields 1,340 to 2,560 tonnes CO₂ per year in the base case. Peat or

coal displacement (0.38 tCO₂ per MWh) yields 1,890 to 3,210 tonnes CO₂ per year in the base case. The full sensitivity matrix is shown in Table 3.

Table 3. Lifecycle CO₂ displacement sensitivity (1.7 MW container, Nordic eight-month heating season).

Scenario	Utilisation	Effective MWh	Gas (0.20 tCO ₂ /MWh)	Oil (0.27 tCO ₂ /MWh)	Peat (0.38 tCO ₂ /MWh)
Low	50%	4,964	993 tCO ₂	1,340 tCO ₂	1,886 tCO ₂
Base low	70%	6,950	1,390 tCO ₂	1,876 tCO ₂	2,641 tCO ₂
Base high	85%	8,439	1,688 tCO ₂	2,279 tCO ₂	3,207 tCO ₂
Ideal	95%	9,432	1,886 tCO ₂	2,547 tCO ₂	3,584 tCO ₂

Net climate benefit is the difference between displaced heating emissions and mining electricity emissions (~0.29 tCO₂ per MWh at the Cambridge survey 52.4 percent sustainable mix). For the 1.7 MW container at base-case utilisation, mining electricity contributes approximately 4,322 tCO₂ per year of generation-side emissions. The net displacement therefore turns clearly positive only when the displaced fuel is peat or oil, or when the local electricity mix is meaningfully cleaner than the global average. The Microsoft and Fortum partnership at Espoo, where the local Finnish grid is more than 90 percent low-carbon, reports a net 400,000 tCO₂ per year savings projection at 350 MW combined thermal output (1,143 tCO₂ per MW per year), consistent with the order of magnitude of the per-container model. The Microsoft and Fortum figure is a vendor projection drawn from a similar accounting basis, not an independently audited measurement; it is offered as a parallel point estimate rather than an independent validation of the per-container model.

4.2.2. Embodied emissions of the cooling fluid

The lifecycle accounting must also include the embodied CO₂ of the cooling fluid itself. Synthetic esters and engineered hydrocarbon dielectric fluids carry an embodied burden of approximately 2 to 4 kg CO₂ per kg fluid based on cradle-to-gate life-cycle inventories of analogous ester and hydrocarbon products [61]. A 1.7 MW container with a fluid charge of approximately 8,000 to 12,000 L (density 0.8 to 0.9 kg per L) carries a fluid mass of 6,400 to 10,800 kg, corresponding to embodied emissions of 13 to 43 tonnes CO₂. Replacement at 5 to 8 year intervals adds the same embodied burden each cycle. Both figures are small relative to the 1,400 to 3,200 tCO₂ per year displacement base case, recovered within the first 1 to 3 weeks of operation, but should be reported in any comprehensive lifecycle assessment.

For two-phase fluorocarbon fluids the embodied burden per kg is several times higher and global-warming-potential leak emissions can be substantial; HFO and HFE alternatives generally have lower per-kg embodied emissions but require recharge after leak events that are more frequent owing to the lower-viscosity working fluids. A full two-phase fluid lifecycle accounting is outside the scope of this review.

4.3. Scalability and geographic constraints

Finland, Sweden, Denmark, and Norway deliver more than 50 percent of heating through district networks. Compute heat integration works in these regions because of cold climate (8 to 10 months annual demand), existing pipe infrastructure, supportive municipal policy, and high conventional energy costs [4]. Many Finnish district-heating systems already source heat from co-generation plants; in these systems, the marginal climate benefit of mining heat is reduced because the displaced source was already efficient [37]. The strongest case exists where networks depend on standalone fossil boilers, which is why the Mäntsälä Yandex installation [38] reports outsized gas-utility CO₂ reductions (40 percent), and why the Finnish district-heating sector has been an early-adopter market for this class of integration.

Germany's EnEFG implementation creates a large near-term market: approximately 1,000 data centres above the 300 kW threshold are subject to the 10/15/20 percent heat-reuse trajectory beginning in 2026 [41, 42]. Successful deployments depend on alignment between data-centre operators, existing district-heating infrastructure, and heat demand in local communities. The Frankfurt FRANKY case demonstrates that even a 2 percent utilisation of waste heat from a single colocation campus can supply more than 60 percent of a 3,000-resident neighbourhood's heating demand [55].

District-heating networks are designed for 30-year lifespans. Mining hardware is obsolete in 3 to 5 years; data-centre hardware in 7 to 10. Municipalities evaluating mining-heated districts must plan for multiple hardware refresh cycles per infrastructure lifecycle [18].

Bitcoin mining provides rapid demand response on a scale that other large-scale compute workloads do not match. Riot Platforms' 700 MW Texas facility curtailed more than 95 percent of power usage during the August 2023 heat wave, earning USD 31.7 million in ERCOT curtailment credits in a single month [S1, S2]. ERCOT large-load applications reached 226 GW in 2025, with AI companies comprising 73 percent of new applications. The demand-response dimension is treated in detail in the Supplementary Material (Section S1). The empirical claim that mining provides demand-response capacity at a scale unmatched by AI workloads is supported by published ERCOT curtailment data; a comparative analysis of demand-response performance across compute workloads is outside the scope of this materials-focused review.

4.4. Asian deployments and disclosure asymmetry

China's data-centre liquid-cooling market reached USD 2.37 billion in 2024 (67 percent year-over-year growth) [43]. Alibaba's single-phase immersion-cooling deployments at scale have been documented to reduce energy consumption by over 35 percent compared to traditional air-cooled data centres [62]. Tencent's Tianjin facility uses a magnetic-suspension heat pump to recover waste heat for residential heating [60]. National policy under the Ministry of Industry and Information Technology's Three-Year Action Plan for New Data Center Development (2021 to 2023) and successor programmes mandates a PUE under 1.3 for newly built large-scale data centres, with a national goal of 1.25 by 2025 [63]. Chinese district heating, however, remains 85 percent powered by coal and natural gas, with less than 4 percent currently sourced from data centres [64], indicating that the

Chinese market is in an earlier phase of the integration trajectory than the Nordic and German markets even at substantially larger scale.

Disclosure asymmetry between Chinese and European data-centre operators makes deployment-level case-by-case validation difficult for the present review. Trade-press reports indicate operational immersion-cooled installations at multiple Alibaba and Tencent sites, but technical disclosure typically does not include the per-site thermal-output and output-temperature data required to populate Table 2. Future reviews on this topic will benefit from improved disclosure norms in the Asian market.

5. Second-Generation Materials: Gallium-Based Liquid Metal Systems

5.1. Thermal performance

Pure gallium exhibits a thermal conductivity of 33.7 W per metre kelvin in the solid state and 24 W per metre kelvin as a liquid, with a latent heat of 80,300 J per kg at a melting point of 29.8 °C [19]. GaInSn eutectic alloys (Galinstan: 68.5 percent Ga, 21.5 percent In, 10 percent Sn) achieve thermal conductivities of 26 to 39 W per metre kelvin as thermal interface materials, with optimised formulations reaching higher values depending on composition, bond-line thickness, surface roughness, and contact pressure [20, 21]. (Galinstan is a registered trademark of Geratherm Medical AG; first use is footnoted.)

It is essential to distinguish two applications of gallium-based materials with very different engineering requirements and readiness levels. Ga-based *thermal interface materials* (TIMs), applied at the die-to-heat-sink junction in milligram quantities, are commercially mature: Sony’s PlayStation 5 uses a Galinstan-based TIM in mass production, and Indium Corporation’s GalliTHERM product line targets HPC semiconductor applications (TRL 8 to 9). Ga-based *cooling loops*, circulating kilograms of liquid metal through metres of facility piping, are essentially undemonstrated at scale (TRL 3 to 4). Bridging this gap is the central scaling challenge for the second generation. The TRL trajectory is summarised in Table 4.

Table 4. Gallium-cooling TRL trajectory.

Application	Current TRL	Representative example	Key remaining barrier	Estimated TRL by 2030
TIM at die-to-heatsink interface	8–9	Sony PS5 (Galinstan); Indium GalliTHERM	Cost reduction; long-term reliability under CTE cycling	9 (mass production extends)
Lab-scale cooling loop (single-rack)	5–6	Beijing Institute of Technology rack prototypes (2023)	Containment redundancy; pump reliability	6–7
Pilot-scale cooling loop (multi-rack)	3–4	DARPA-funded prototypes	Material compatibility; leak detection; supply chain	5–6

Facility-scale cooling loop (>1 MW)	2–3	None demonstrated	All of the above plus economics	4–5
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5.2. Phase-change properties and figures of merit

Boteler et al. demonstrated that for a temperature rise of 20 °C, gallium absorbs seven times more energy per unit volume than copper and three times more than organic paraffin PCMs [19]. Shao et al. developed a figure-of-merit for PCM cooling capacity; metallic alloys achieve FOMs one to two orders of magnitude higher than organic PCMs ($2,603 \times 10^3 \text{ W}^2 \text{ s K}^{-1} \text{ m}^{-4}$ for a Bi-based metallic alloy versus 22×10^3 for paraffin wax) [22].

5.3. Experimental validation under pulsed thermal loads

Gonzalez-Nino et al. demonstrated that metallic PCMs reduced temperature rise by 60 to 68 percent under 19 ms pulsed thermal loads at heat fluxes up to 338 W per square centimetre, compared to only 14 to 17 percent for the best organic PCM [23]. At 160 W pulse power, the metallic PCM maintained temperature within 5 °C of its melting point while the dielectric baseline rose 119 °C. *Important context:* the Gonzalez-Nino experiments tested 19 ms transient pulses, characteristic of switching transients in power electronics. Steady-state thermal loads at facility scale (continuous heat dissipation over hours) require additional analysis of melting-front propagation, density-driven convection, and supercooling-induced solidification anomalies (Section 6.4); the 60 to 68 percent reduction is therefore an upper bound on the steady-state benefit. Translation of pulsed-load performance to AI training workloads with multi-second to minute-scale thermal envelopes is an active research area.

Ma and Liu demonstrated thermal-conductivity enhancement ratios up to 2.3× with 20 percent CNT volume fraction in liquid gallium [24]. This result, while promising, has not been reproduced at scale; CNT dispersion in liquid gallium faces sedimentation, agglomeration, and oxide-entrapment challenges at volumes beyond the laboratory. The liquid-metal TIM market reached USD 305.9 million in 2025, projected to USD 479.7 million by 2032 [25]. Boteler et al.’s time-scale-matched PCM systems, pairing metallic PCMs with organic PCMs, demonstrated 25 to 50 percent mass reduction relative to pure metallic systems [19]; that demonstration was performed on milligram quantities and has not been scaled to facility loops.

6. Engineering Challenges for Gallium-Based Cooling Systems

6.1. Corrosion and material compatibility

Gallium alloys aggressively attack aluminium via liquid-metal embrittlement. Deng and Liu found that 6063 aluminium alloy suffered complete gallium infiltration within five hours at 120 °C [29]. T2 copper exhibited slow corrosion at approximately 30 to 60 µm per month at 60 °C. 1Cr18Ni9 stainless steel showed zero detectable gallium infiltration after 30 days

at 60 °C [29]. The practical materials palette for facility-scale cooling loops is therefore nickel-plated copper or 316L stainless steel throughout.

6.2. Electrical conductivity and containment

Liquid metals are electrically conductive ($\sim 3.5 \times 10^6$ S per metre). A dielectric-fluid leak is a maintenance event; a liquid-metal leak is a catastrophic short-circuit risk. Gallium-based systems require sealed, isolated cooling channels rather than open-bath immersion. This constraint is structural: it means that gallium loops cannot adopt the open-tank architecture of first-generation immersion cooling, and must instead resemble closed-loop cold-plate systems with much tighter leak-detection requirements.

6.3. Surface tension and wettability

Gallium exhibits high surface tension (~ 709 mN per metre vs. ~ 73 mN per metre for water), requiring oxide-layer engineering for uniform thermal contact. Native oxide formation on gallium and Galinstan is rapid (~ 2 nm Ga_2O_3 within seconds of air exposure) and improves wetting on metal substrates by reducing the effective interfacial energy, but the oxide layer also increases interfacial thermal resistance.

6.4. Supercooling mitigation

Bulk gallium can remain liquid to approximately -38 °C, nearly 68 °C below its 29.8 °C melting point, owing to the absence of nucleation sites for solid formation. Zhang et al. demonstrated that 0.5 wt.% TeO_2 particles reduced supercooling by 44 percent [30], offering one route to a more reliable phase-change response. Supercooling is a critical concern for facility-scale loops because uncontrolled solidification could cause local blockages.

6.5. Toxicity considerations

Gallium itself exhibits low acute toxicity. However, Galinstan contains approximately 21.5 percent indium, and indium compounds have documented pulmonary toxicity associated with occupational exposure. The American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Value for indium and its compounds is 0.1 mg per cubic metre as an 8-hour time-weighted average [65]. Indium lung disease (pulmonary alveolar proteinosis) has been documented in workers in indium tin oxide manufacturing and represents the controlling occupational-health concern for any facility-scale gallium-cooling deployment that uses indium-containing alloys [66]. At facility scale, handling protocols must address inhalation risks from aerosolised particles during alloy preparation, leak cleanup, decommissioning, and recycling, requiring local exhaust ventilation and respiratory protection per OSHA 1910.134 (or EU equivalent). These requirements add operating cost and worker-training overhead that are not present in dielectric-fluid systems.

6.6. Implications for heat-reuse systems

A gallium cooling loop could capture waste heat at higher temperatures with lower thermal resistance, potentially enabling output temperatures above 80 °C. If Ga-based systems can

deliver output temperatures consistently above 80 °C, they may eliminate the need for heat-pump boosting that some immersion-cooled and cold-plate installations still require for network compatibility, while remaining within the 60 to 95 °C window targeted by district-heating modernisation programmes. This is a projection, not yet demonstrated in district-heating applications.

6.7. Regulatory exposure

Three regulatory frameworks together govern facility-scale deployment economics for gallium cooling. ECHA's REACH framework regulates indium aerosol exposure and is currently silent on gallium metal but tightening on PFAS (Section 3.1.1). OSHA in the United States and the equivalent EU agencies regulate occupational exposure to indium and indium tin oxide; ACGIH's 0.1 mg per cubic metre TLV is the operative standard at the time of writing [65]. The People's Republic of China's Ministry of Commerce imposed export controls on gallium and germanium effective 1 August 2023 [32], which raised gallium prices internationally and tightened supply for non-Chinese semiconductor and cooling-loop manufacturers. Together, these frameworks govern (i) where a facility-scale gallium loop can be sited, (ii) how it must be maintained, and (iii) where it can source its working fluid. Operators planning facility-scale gallium cooling must integrate compliance design across all three frameworks from project initiation.

7. Materials Supply and Circularity

This section, expanded from the original review's gallium supply-chain coverage, addresses the supply, circularity, and end-of-life considerations across all three generations.

7.1. Gallium supply chain

Gallium is recovered as a by-product of aluminium and zinc refining [26]. Approximately 95 percent of consumption goes to GaAs wafer fabrication. China accounts for approximately 80 percent of primary production and imposed export controls in July 2023 [27, 32]. A lifecycle analysis of global gallium flows (2000 to 2021) found that of 5,862 tonnes entering the industrial cycle, only 967 tonnes remain in active use, with 83.5 percent wasted and 4,446 tonnes in recoverable manufacturing or electronic waste [31]. Facility-scale cooling loops are circulatory systems in which gallium recirculates indefinitely; the operative supply-chain metric is make-up volume for leak and maintenance losses, not total fill volume, which substantially reduces ongoing demand relative to the semiconductor pathway where gallium is permanently embedded in wafers. The supply constraint is manageable at current projected adoption rates but must be monitored given GaN power-electronics demand growth and continuing Chinese export-control evolution.

7.2. Dielectric fluid lifecycle and recycling

Synthetic esters and engineered hydrocarbons used as immersion fluids are recyclable through established lubricant-reprocessing channels, with reclaim yields of 70 to 85

percent typical for closed-loop industrial fluids. End-of-life fluid that has accumulated TAN above 0.5 mg KOH per gram is typically incinerated for energy recovery rather than reclaimed, contributing additional cradle-to-grave embodied emissions. Two-phase fluorocarbon fluids face a different end-of-life challenge: PFAS-classified compounds cannot be incinerated below 1,100 °C without releasing PFAS-class atmospheric pollutants, and dedicated PFAS-destruction infrastructure is limited. The PFAS phase-out (Section 3.1.1) makes circularity questions for two-phase systems acute over the next five to seven years.

7.3. ASIC and GPU end-of-life

ASIC mining hardware is obsolete in 3 to 5 years; AI GPUs in 5 to 8 years; general-purpose data-centre hardware in 7 to 10 years [18]. ASIC obsolescence generates an estimated 15,000 to 30,000 tonnes of e-waste annually globally [67]. Recycling pathways exist for the printed circuit boards (gold, copper, palladium recovery) and for the heat-sinks and chassis (aluminium, copper), but not for the silicon dies themselves, which are typically landfilled or downcycled to refractory aggregate. Cooling-system metals (copper piping, stainless-steel manifolds, nickel plating) are generally recovered at end of life as part of the broader scrap-metal stream. The cooling architecture itself does not change the e-waste profile of the compute hardware, but it does affect recycling logistics: immersion-fluid drainage and decontamination steps add labour per kilogram recovered relative to air-cooled recycling.

7.4. Circularity verdict

Across all three generations, circularity for the cooling material itself ranges from good (single-phase dielectric fluids, recyclable through industrial channels; gallium loops, indefinitely recirculating) to challenging (two-phase fluorocarbon fluids, PFAS-restricted destruction). The dominant lifecycle burden remains the compute hardware, not the cooling material. RSER readers should weigh cooling-material circularity claims in this context.

8. Third-Generation Materials: Embedded Two-Phase Cooling

8.1. The ICECool programme and successors

DARPA's ICECool programme demonstrated chip-level heat-flux removal exceeding 1 kW per square centimetre, with hot-spot cooling approaching 5 kW per square centimetre [28]. IBM demonstrated microchannels in a production microprocessor achieving junction temperatures approximately 25 °C lower than the air-cooled baseline while reducing power by more than 10 W. R1234ze (E) (GWP < 1) can produce exit temperatures in the 60 to 80 °C range suitable for district heating. Microsoft's collaboration with Corintis, announced in September 2025, integrates microfluidic cooling channels directly within AI accelerator silicon and reports up to three times the heat-removal capability of conventional cold plates, targeting commercial deployment in next-generation AI training infrastructure [49]. This represents a significant TRL advance for embedded two-phase

cooling, from DARPA-demonstration scale to commercial-product scale within the past two years.

Manufacturing-scalability questions remain. Through-silicon-via integration yields for microchannel-in-die designs are reportedly above 90 percent at 7 nm and 5 nm nodes for sub-millimetre die sizes, but published yield data at production AI-accelerator die sizes (around 800 mm² for current-generation GPUs) is limited. Microchannel reliability under thermal cycling has been demonstrated above 1,000 cycles at temperature swings of 70 °C without measurable degradation in DARPA ICECool reports, but the 100,000-cycle service-life target for facility deployment has not been tested publicly. Refrigerant-loop reliability at facility scale (rather than at chip-package or rack scale) introduces additional challenges in fluid-distribution manifold design, in particular thermal-expansion management across long runs and pressure-drop balancing across heterogeneous workloads. The path from the Microsoft and Corintis chip-scale demonstration to integrated facility-scale embedded two-phase cooling is the central engineering question for the third generation.

8.2. Comparative materials in the design space

Several materials sit alongside the three primary generations in the materials design space (Figure 5).

CVD diamond achieves thermal conductivities of 1,700 to 2,000 W per metre kelvin, making it the highest-conductivity bulk material currently producible at engineering scale. It manages hotspots at the transistor level via heat-spreader integration. CVD diamond and gallium liquid metals are complementary, not competitive: diamond addresses the chip-to-package interface; gallium addresses the package-to-loop and loop-to-facility transport. CVD-diamond integration into commercial GaN power electronics has reached TRL 7 to 8 with deployments in the SAFE-Diamond programme.

Hexagonal boron nitride (hBN) offers theoretical parity with diamond (theoretical thermal conductivity ~2,000 W per metre kelvin in plane) plus electrical insulation, but practical composites achieve only 5 to 15 W per metre kelvin owing to interface scattering and orientation distribution. The manufacturing gap for high-thermal-conductivity hBN composites is approximately a decade at current research rates. TRL 4 to 5 for chip-scale composites; TRL 2 to 3 for facility-scale heat-spreader applications.

Graphene nanofluids achieve thermal conductivities of approximately 1 to 2 W per metre kelvin, a meaningful improvement over dielectric fluids but 15 to 30 times below gallium. Stability of graphene-based suspensions over the 5,000 to 10,000 hour service intervals required for facility-scale cooling loops has not been demonstrated. TRL 3 to 4 at the time of writing.

Together with the three primary generations, these alternative materials trace the materials-design space mapped in Figure 5: thermal conductivity (W per metre kelvin) plotted against operating length scale (metres). Each material class occupies a distinct region; no single material spans the full thermal chain from transistor to district-heating grid.

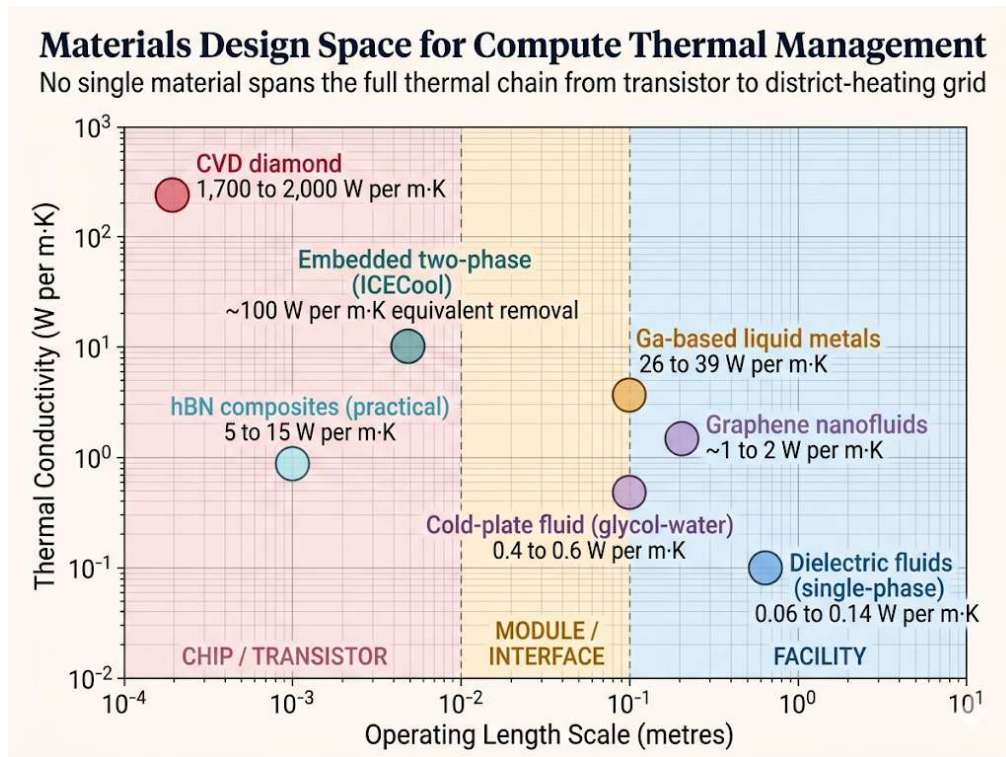


Figure 5. Materials Design Space for Compute Thermal Management. Thermal conductivity (W per metre kelvin) plotted against operating length scale (metres). Each material class occupies a distinct region of the design space: CVD diamond at the chip and transistor scale (1,700 to 2,000 W per metre kelvin); embedded two-phase ICECool systems at the chip-to-module scale; gallium-based liquid metals at the module-to-loop scale (26 to 39 W per metre kelvin); hexagonal boron nitride composites at the chip-to-module scale (5 to 15 W per metre kelvin practical); graphene nanofluids at the loop scale (1 to 2 W per metre kelvin); and dielectric fluids at the facility scale (0.06 to 0.14 W per metre kelvin). No single material spans the full thermal chain from transistor to district-heating grid.

9. Conclusion

Three regulatory and market shifts now make the materials-design framing developed in this review consequential beyond the laboratory. The European Union's Energy Efficiency Directive (2023/1791) mandates waste-heat recovery from new data centres above 1 MW, and Germany's EnEfG sets a binding 10/15/20 percent heat-reuse trajectory for 2026 to 2028. AI data-centre electricity demand is projected to exceed 1,000 TWh by the end of 2026, more than seven times Bitcoin mining's consumption. ERCOT large-load applications reached 226 GW in 2025 with AI now accounting for 73 percent of new requests. Within these constraints, the cooling material is the design variable that determines whether compute waste heat becomes a usable energy product or a thermodynamic loss.

First-generation materials (dielectric immersion fluids and direct-to-chip cold plates) enable the temperature-gradient breakthrough: raising waste-heat output from 25 to 35 °C

in air-cooled facilities to 50 to 75 °C in liquid-cooled facilities, crossing or approaching the district-heating threshold. Seventeen operational deployments across nine countries validate this generation, with verified emissions data available for the AWS Dublin (704 tCO₂ abated, 2024), Telehouse Frankfurt FRANKY (~400 tCO₂ saved), and Yandex Mäntsälä (40 percent gas-utility CO₂ reduction) installations, and a 400,000 tCO₂ per year projection for the Microsoft and Fortum partnership at Espoo. The lifecycle CO₂ displacement model presented here yields net displacement of approximately 1,400 to 3,200 tonnes CO₂ per 1.7 MW container per year at base-case utilisation.

Second-generation gallium-based liquid metals (26 to 39 W per metre kelvin at the thermal interface) offer 200 to 600 times the conductivity of dielectric fluids, with 60 to 68 percent temperature-rise reduction under pulsed loads. Mature TIM applications (TRL 8 to 9) must be clearly distinguished from undemonstrated facility-scale cooling loops (TRL 3 to 4). Engineering challenges in corrosion, electrical conductivity, indium occupational toxicity, and supply-chain concentration are tractable but require integrated design-to-deployment compliance. Third-generation embedded two-phase microfluidic cooling, demonstrated at over 1 kW per square centimetre under DARPA's ICECool programme and now reaching commercial scale through the 2025 Microsoft and Corintis collaboration, points toward chip-level thermal management at heat-flux densities that facility-level immersion cannot reach.

Three open research questions will shape the next decade of work in this area. First, can gallium-based facility-scale cooling loops be demonstrated at TRL 6 to 7 within a five-year horizon, or do the engineering challenges (containment, electrical safety, supply chain) lock gallium into the TIM-and-prototype niche? Second, can embedded two-phase microfluidic cooling reach manufacturing yields and reliability metrics that justify integration into production AI accelerators at hyperscale, beyond the Microsoft and Corintis pilot? Third, can district-heating tariff structures evolve to compensate compute operators for delivered heat at scale, beyond the open-tariff model pioneered by Stockholm Exergi, so that heat reuse becomes a revenue line rather than a regulatory cost?

The answer to all three will be material-dependent. The materials choice is not merely a thermal-engineering decision but a sustainable energy-systems design decision, with downstream consequences for water consumption, carbon displacement, and district-heating compatibility. The materials-design frontier documented here is the engineering path along which compute infrastructure will continue to shed its thermal waste profile and integrate into the broader sustainable energy system.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author holds no material positions in Bitcoin, mining companies, or energy firms discussed herein.

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Use of AI-Assisted Tools

The author used Anthropic Claude (Opus and Sonnet variants) for editorial revision (sentence-level rewriting for clarity and concision), structural reorganisation (section sequencing, heading hierarchy, table formatting), figure-production assistance (Python plot scripting; figure-caption drafting), and manuscript formatting (Markdown and LaTeX conversion; reference list formatting). All factual claims, numerical values, citations, the three-generation taxonomy, the lifecycle CO₂ displacement model, the materials-property comparisons, and the analytical conclusions are the author's work. No AI tool generated factual claims, citations, or numerical values that were not subsequently verified by the author against the primary sources. The author used WebSearch and Google Scholar for primary literature retrieval; the AI tools assisted with search-query formulation but did not replace the author's literature assessment. Disclosure language follows RSER's 2025 guidance on generative AI use in submitted manuscripts.

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Note on literature scope: The author does not claim the reference set is exhaustive. Readers should consult Deng and Liu, *Liquid Metals for Advanced Energy Applications* (AIP, 2022) and *Advanced Liquid Metal Cooling* (Springer, 2024) for additional references on second-generation materials. For data-centre waste-heat recovery and district-heating integration,

Yuan et al. (RSER 2023, 2025) and Wahlroos et al. (RSER 2018) provide the comprehensive system-side complement to the present materials-side review.

Supplementary Material

S1. Demand response and grid integration. ERCOT large-load curtailment data (Riot Platforms 700 MW curtailment August 2023, USD 31.7M earned), 2025 ERCOT large-load application volumes (226 GW total, 73 percent AI), comparative analysis of demand-response performance across compute workloads, and discussion of grid-services revenue as an additional sustainability lever for compute facilities. (Provided as separate file Supplementary_Material_REVISED.docx.)